UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the May/June 2006 question paper

5070 CHEMISTRY

5070/03 **Paper 3** **maximum raw mark 40**

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. It shows the basis on which Examiners were initially instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began. Any substantial changes to the mark scheme that arose from these discussions will be recorded in the published *Report on the Examination*.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the *Report on the Examination*.

- CIE will not enter into discussion or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the May/June 2006 question papers for most IGCSE and GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses

UNIVERSITY of CAMBRIDGE
International Examinations

1 16 marks

(a) titration **[12]**

4 marks for each of two titration results within 0.2 cm^3 of the Supervisor's value.
3 marks for results within 0.3 cm^3 etc.
no marks for results more than 0.5 cm^3 from the supervisor's value

maximum of 3 marks for concordance, i.e. results within 0.2 cm^3

1 mark for taking a correct average

(b) concentration of **X** **[2]**

method (1) answer (1)

(c) M_r of **X** **[2]**

method (1) answer (1)

2 24 marks

Solid R (manganese(IV) oxide)

Test 1	effervesces	**[1]**
	gas relights a glowing splint	**[1]**
	oxygen produced	**[1]**
	exothermic reaction	**[1]**
	solid is insoluble (or appears unchanged)	**[1]**

Test 2	effervesces	**[1]**
	gas bleaches litmus	**[1]**
	chlorine produced	**[1]**
	brown solution	**[1]**

Solution S (manganese(II) sulphate)

Test 3	beige ppt/pale brown	**[2]**
	insoluble in excess	**[1]**
	darkens on the surface	**[1]**
	brown ppt	**[1]**
	effervesces	**[1]**
	gas relights a glowing splint or oxygen produced	**[1]**

Test 4	no reaction	**[1]**
	no reaction	**[1]**

Test 5	white ppt	**[1]**
	insoluble in acid	**[1]**

Test 6	brown ppt	**[1]**

© University of Cambridge International Examinations 2006

Conclusion

In Test 1 R is a catalyst **[1]**

In Test 2 R is an oxidising agent **[1]**

The anion is a sulphate **[1]**

Any 24 marks to score.

© University of Cambridge International Examinations 2006